

Answer **all** the questions in the spaces provided.

1 (a) (i) Define *impulse*.

Impulse on object Y is the product of the force acting on object Y and the time of impact of the force.

[1]

(ii) State the principle of *conservation of linear momentum*.

The principle of conservation of linear momentum states that the total momentum of the bodies remain constant [B1] if there is no resultant/net external forces acting on it. [B1]

[2]

(b) Two isolated objects, X and Y travel along the same straight line with speeds 3.5 m s^{-1} and 2.0 m s^{-1} respectively as shown in Fig. 1.1. The objects collide elastically and continue to